

Evidence Report

Name: Tytus Gonzalez

Title: FAT32 USB Image Examination

Case: 26-T006

Date: 04/23/2026

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Document Revision History

Name	Revision Date	Version	Description
Tytus Gonzalez	2026-04-23	0.1	Draft
Tytus Gonzalez	2026-04-24	1.0	Final

Executive Summary

On April 9, 2025, a forensic investigation was conducted on a USB thumb drive belonging to Ms. Jessica Langford following her termination from Acme Corp. and subsequent filing of a whistleblower and wrongful termination lawsuit. The purpose of this investigation was to determine whether Ms. Langford exfiltrated sensitive company data after her termination and to identify any evidence related to her motives or methods. I, Tytus Gonzalez, performed the analysis using a Windows forensic virtual machine, utilizing both The Sleuth Kit (TSK) and WinHex to examine the disk image in a controlled environment.

The investigation began with identifying the disk structure using TSK. The `mmls` command revealed that the disk utilized a GUID Partition Table (GPT) partitioning scheme with a single primary partition beginning at sector 2048. This finding was verified in WinHex by observing the "EFI PART" signature and confirming the presence of a protective MBR signature (0x55AA). The file system was then analyzed using `fsstat`, which identified the partition as FAT32 with the volume label "LANGFORDUSB." Key filesystem parameters, including sector size, reserved sectors, and FAT layout, were extracted to calculate the root directory location. The root directory offset was computed and verified in WinHex, where valid directory entries and readable file names were observed.

Further analysis was conducted using the `fls` command to enumerate files and directories. The root directory contained several regular files, including *Vendor_Contract_MercuryLLC.docx*, *Lawsuit_Docs_Settlement_Draft.docx*, and *Compliance_Audit_Internal_Review.docx*. Multiple directories, including *Contracts*, *Financials*, *Legal*, and *Notes*, were identified as deleted, indicating potential attempts to conceal data organization. A recursive file listing revealed numerous deleted files within these directories, including *password_list.txt* and *TO_DO_BEFORE_TERMINATION.txt*, as well as financial, legal, and internal audit documents.

The recovery and examination of these files indicate that Ms. Langford had access to sensitive corporate information and may have prepared for her termination by organizing and retaining critical documents. Notably, the presence of *password_list.txt* suggests that credentials were stored on the device, which may assist in accessing encrypted systems or

accounts. The combination of deleted directories and recoverable files suggests an attempt to obscure evidence rather than securely erase it.

This investigation demonstrates a comprehensive forensic analysis process, including disk structure identification, file system analysis, directory and file enumeration, recovery of deleted artifacts, and verification using both automated tools and manual hex analysis. The findings provide strong evidence of potential data exfiltration and premeditated actions by Ms. Langford, supporting further investigative and legal proceedings.

Synopsis

The Professor provided the following questions.

1. Using either a hex editor or TSK, determine the disk partitioning type (MBR or GPT).
2. Using either a hex editor or TSK, how many partitions are on the disk?
3. What is the file system for the first partition?
4. What is the volume label (boot sector or root directory) for this partition?
5. What is the hexadecimal offset for the first partitions root directory? This will require some calculation and verification.
6. What are the first eleven characters of the root directory? Do not give hexadecimal characters and should look familiar.
7. There are three regular files under the root directory of the file system, what are they?
8. Are there any directories under the root directory? Name them.
9. Are these folders allocated?
10. List all unallocated directories?
11. List all unallocated files?
12. Are any unallocated files you just mentioned located elsewhere on this disk? List them.
13. Are there any files that may give us insight into the motives or methods of Ms. Langford? List the file names?
14. What is the content of this file?
15. Another team is reporting that Ms. Lanford encrypted her laptop with a password. Can you give them any information that would help their investigation?

Evidence Collected

The section outlines all physical evidence items collected

Name	Identifier (Tag)	Description
Usb-whistleblower.img	Usb-whistleblower	DD image of USB device provided by instructor

This section provides details of the digital evidence collected

Name	Usb-whistleblower.img
Type:	DOS/MBR boot sector; partition 1
Size:	3459738368 bytes
MD5:	850532AE1CA15B95E141EABE8004695A
SHA1:	0CBBB41DCA93351C4D95A597658FD3FC4B7E6375
SHA256	0968EA1DF2F80CA5CFB59611ABE9D4EAC5FD49E8127017AEF0306CFA61FA3AE9

Tools Used

The following outlines the workstations and all hardware and software used for this investigation.

Workstation

Hostname	Operating System	Build	Physical Virtual	/	Built
Windows 11x64_DFIR	Windows 11 Home	2021	Virtual		02/06/2026

Software

Name	Version	Release	Purpose
WinHex/X-ways Forensics	21.6	Oct 19, 2025	A Universal hexadecimal editor, particularly helpful in the realm of computer forensics, data recovery, low-level data processing, and IT security
The Sleuth kit	4.14.0	April 14, 2025	A forensic toolset used to analyze disk images, recover files, and examine file system metadata in digital investigations.

Case Information & Observables

This case involved collecting, processing, and analyzing a Windows-based laptop provided by the Client. The user assigned to this laptop is “Road Runner,” with the username beepbeep@acme.corp (OMIT HYPERLINKS). The following are my observations and responses to Client questions.

Observables

1. Using either a hex editor or TSK, determine the disk partitioning type (MBR or GPT).

The disk uses a GUID Partition Table (GPT) partitioning scheme because ‘mmls’ identifies a ‘GPT Header’ and the WinHex view shows the ‘EFI PART’ signature.

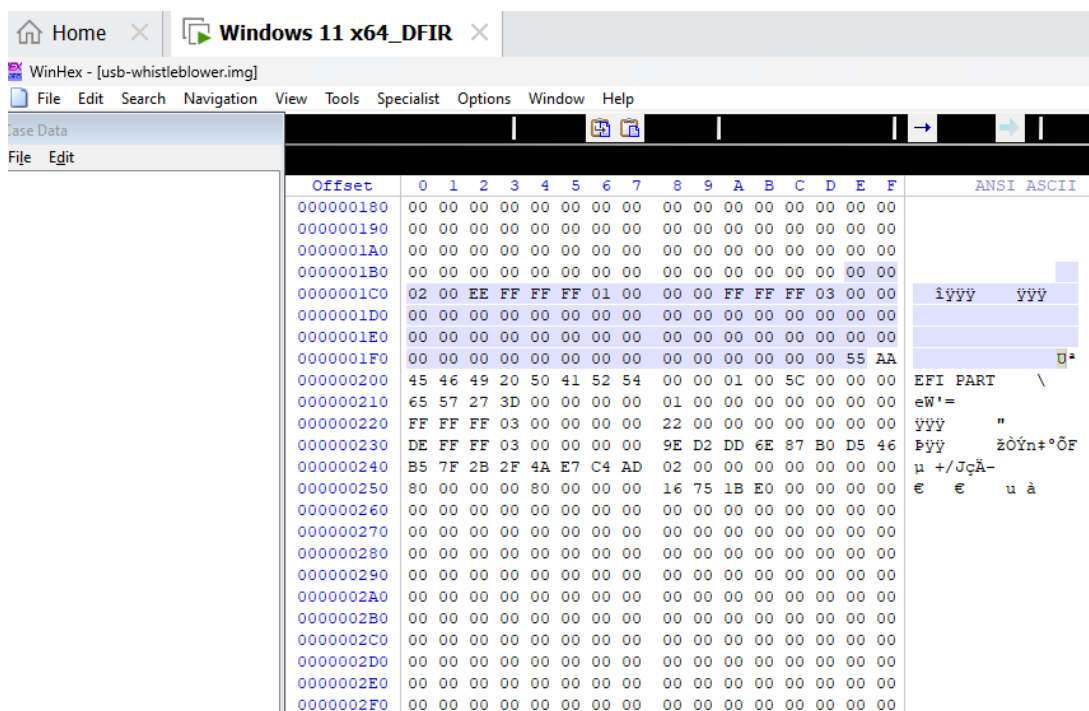


Figure 1: WinHex view of the partition table

```
Administrator: Windows PowerShell

Administrator in E:\usb-whistleblower.img
cd..

Administrator in E:\
Get-FileHash .\usb-whistleblower.img -Algorithm SHA256

Administrator in E:\
mmls usb-whistleblower.img
Invalid magic value (raw_open: image "usb-whistleblower.img" - is a directory)

Administrator in E:\
cd usb-whistleblower.img

Administrator in E:\usb-whistleblower.img
mmls usb-whistleblower.img
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length    Description
000: Meta     0000000000 0000000000 0000000001 Safety Table
001: -----  0000000000 0000002047 0000002048 Unallocated
002: Meta     0000000001 0000000001 0000000001 GPT Header
003: Meta     0000000002 0000000033 0000000032 Partition Table
004: 000      0000002048 0000204799 0000202752 primary
005: -----  0000204800 0067108863 0066904064 Unallocated

Administrator in E:\usb-whistleblower.img
```

Figure 2: TSK 'mmls' command to find partition type

2. Using either a hex editor or TSK, how many partitions are on the disk?

There is one primary partition on the disk because the 'mmls' output lists a single primary partition at sector 2048.

```
Administrator: Windows PowerShell

Administrator in E:\usb-whistleblower.img
cd..

Administrator in E:\
Get-FileHash .\usb-whistleblower.img -Algorithm SHA256

Administrator in E:\
mmls usb-whistleblower.img
Invalid magic value (raw_open: image "usb-whistleblower.img" - is a directory)

Administrator in E:\
cd usb-whistleblower.img

Administrator in E:\usb-whistleblower.img
mmls usb-whistleblower.img
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

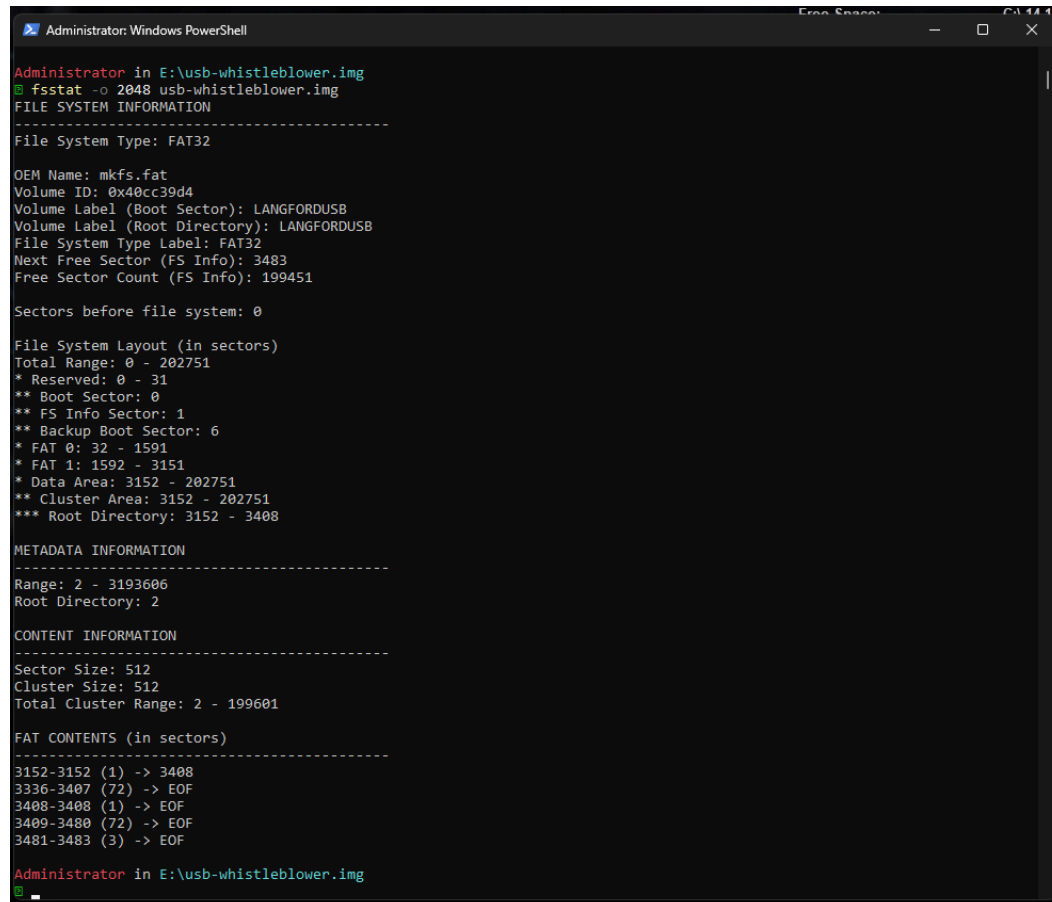
    Slot      Start      End      Length    Description
000: Meta     0000000000 0000000000 0000000001 Safety Table
001: -----  0000000000 0000002047 0000002048 Unallocated
002: Meta     0000000001 0000000001 0000000001 GPT Header
003: Meta     0000000002 0000000033 0000000032 Partition Table
004: 000      0000002048 0000204799 0000202752 primary
005: -----  0000204800 0067108863 0066904064 Unallocated

Administrator in E:\usb-whistleblower.img
```

Figure 3: TSK 'mmls' command shows partition table

3. What is the file system for the first partition?

The file system is FAT32 and was confirmed using TSK as well as WinHex. Using 'fsstat' it directly reports the file system, and the boot sector (2048) in WinHex confirms this as well.



```
Administrator in E:\usb-whistleblower.img
fsstat -o 2048 usb-whistleblower.img
FILE SYSTEM INFORMATION
-----
File System Type: FAT32

OEM Name: mkfs.fat
Volume ID: 0x40cc39d4
Volume Label (Boot Sector): LANGFORDUSB
Volume Label (Root Directory): LANGFORDUSB
File System Type Label: FAT32
Next Free Sector (FS Info): 3483
Free Sector Count (FS Info): 199451

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 202751
* Reserved: 0 - 31
** Boot Sector: 0
** FS Info Sector: 1
** Backup Boot Sector: 6
* FAT 0: 32 - 1501
* FAT 1: 1592 - 3151
* Data Area: 3152 - 202751
** Cluster Area: 3152 - 202751
*** Root Directory: 3152 - 3408

METADATA INFORMATION
-----
Range: 2 - 3193606
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 199601

FAT CONTENTS (in sectors)
-----
3152-3152 (1) -> 3408
3336-3407 (72) -> EOF
3408-3408 (1) -> EOF
3409-3480 (72) -> EOF
3481-3483 (3) -> EOF

Administrator in E:\usb-whistleblower.img
```

Figure 4: TSK 'fsstat' command used to find file system type

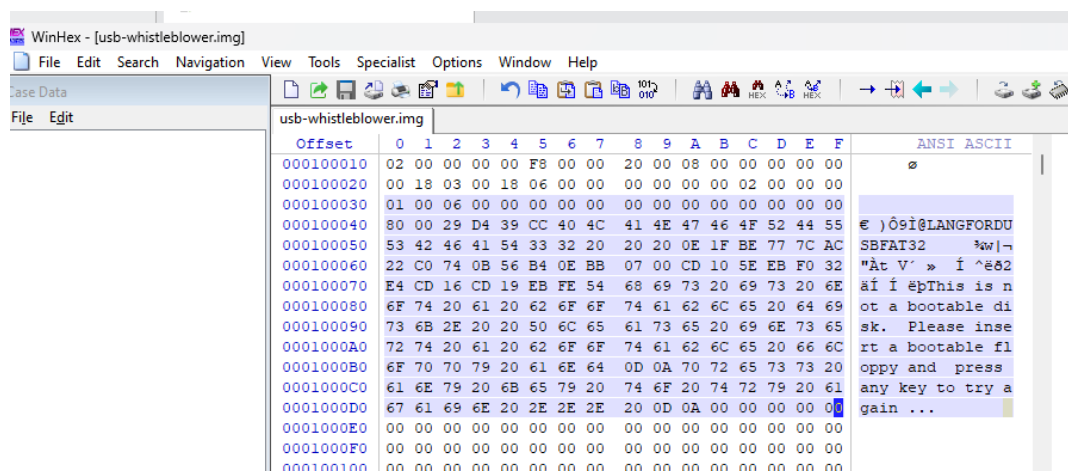
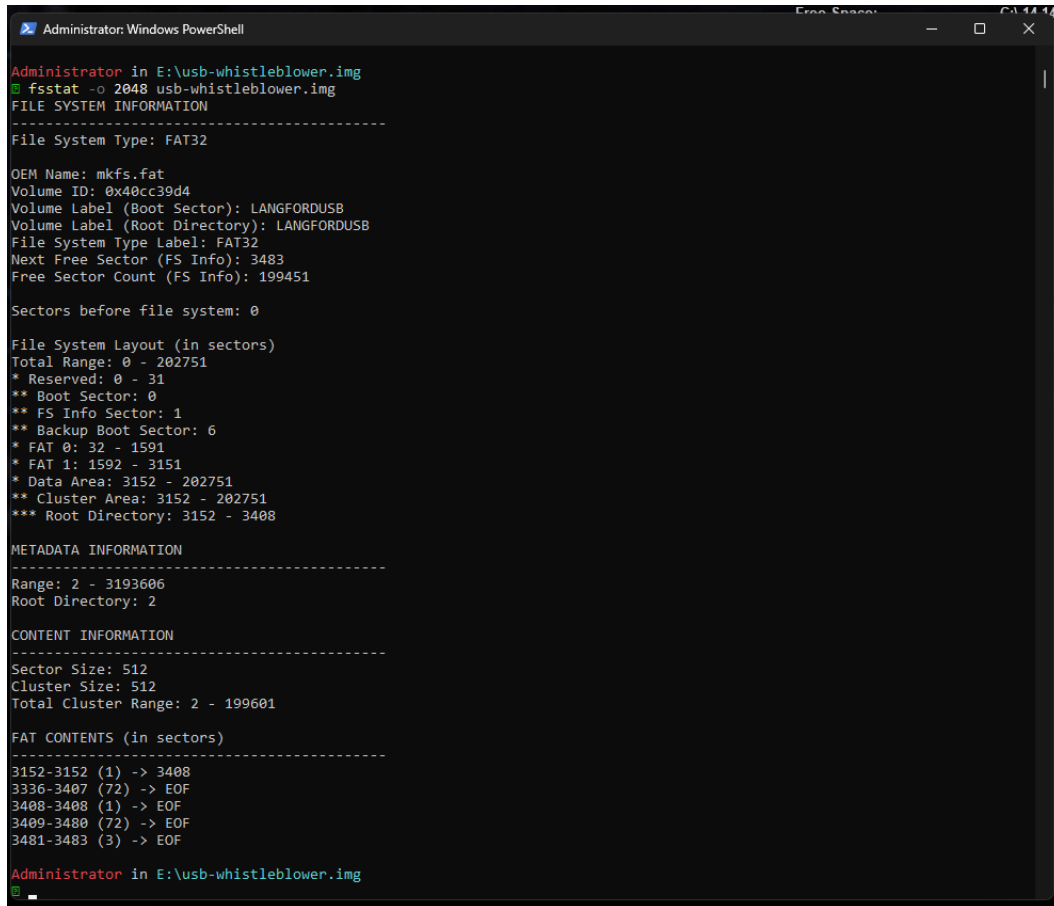


Figure 5: WinHex boot sector to find file type

4. What is the volume label (boot sector or root directory) for this partition?

The volume label is identified as 'LANGFORDUSB'. The volume label was taken from fsstat as well as WinHex in the boot sector.



```
Administrator: Windows PowerShell
Administrator in E:\usb-whistleblower.img
fsstat -o 2048 usb-whistleblower.img
FILE SYSTEM INFORMATION
-----
File System Type: FAT32

OEM Name: mkfs.fat
Volume ID: 0x40cc39d4
Volume Label (Boot Sector): LANGFORDUSB
Volume Label (Root Directory): LANGFORDUSB
File System Type Label: FAT32
Next Free Sector (FS Info): 3483
Free Sector Count (FS Info): 199451

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 202751
* Reserved: 0 - 31
** Boot Sector: 0
** FS Info Sector: 1
** Backup Boot Sector: 6
* FAT 0: 32 - 1591
* FAT 1: 1592 - 3151
* Data Area: 3152 - 202751
** Cluster Area: 3152 - 202751
*** Root Directory: 3152 - 3408

METADATA INFORMATION
-----
Range: 2 - 3193606
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 199601

FAT CONTENTS (in sectors)
-----
3152-3152 (1) -> 3408
3336-3407 (72) -> EOF
3408-3408 (1) -> EOF
3409-3480 (72) -> EOF
3481-3483 (3) -> EOF

Administrator in E:\usb-whistleblower.img
```

Figure 6: TSK 'fsstat' command used to find volume label

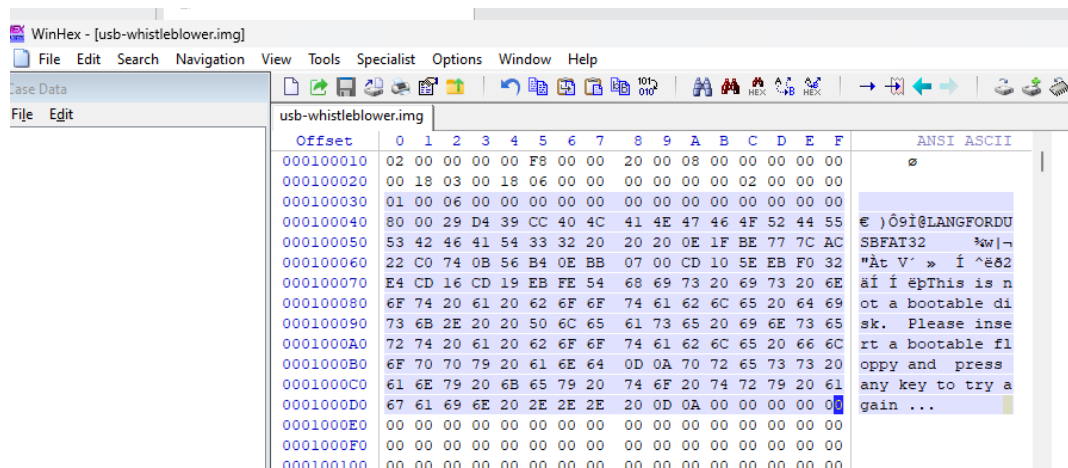
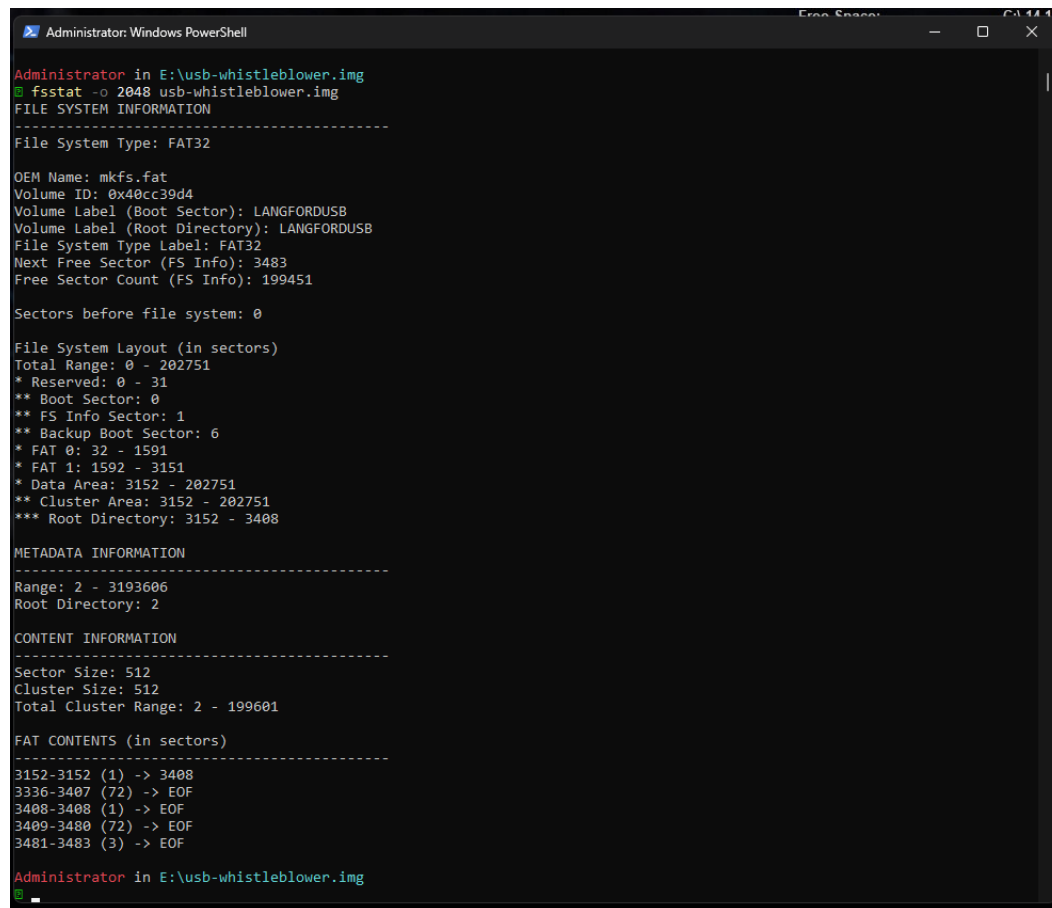


Figure 7: WinHex boot sector to find volume label

5. What is the hexadecimal offset for the first partitions root directory? This will require some calculation and verification.

The hexadecimal offset of the root directory is 0x0028A000 because the root directory begins at sector 3152 relative to the partition. Then when you add the sector to the partition start (2048 + 3152) you get 5200. Then when you multiply by 512 bytes per sector the result is 0x0028A000.



```
Administrator: Windows PowerShell
Administrator in E:\usb-whistleblower.img
fsstat -o 2048 usb-whistleblower.img
FILE SYSTEM INFORMATION
-----
File System Type: FAT32

OEM Name: mkfs.fat
Volume ID: 0x40cc39d4
Volume Label (Boot Sector): LANGFORDUSB
Volume Label (Root Directory): LANGFORDUSB
File System Type Label: FAT32
Next Free Sector (FS Info): 3483
Free Sector Count (FS Info): 199451

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 202751
* Reserved: 0 - 31
** Boot Sector: 0
** FS Info Sector: 1
** Backup Boot Sector: 6
* FAT 0: 32 - 1591
* FAT 1: 1592 - 3151
* Data Area: 3152 - 202751
** Cluster Area: 3152 - 202751
*** Root Directory: 3152 - 3408

METADATA INFORMATION
-----
Range: 2 - 3193606
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 199601

FAT CONTENTS (in sectors)
-----
3152-3152 (1) -> 3408
3336-3407 (72) -> EOF
3408-3408 (1) -> EOF
3409-3409 (72) -> EOF
3481-3483 (3) -> EOF

Administrator in E:\usb-whistleblower.img
```

Figure 8: TSK 'fsstat' command to find the root directory sector

6. What are the first eleven characters of the root directory? Do not give hexadecimal characters and should look familiar.

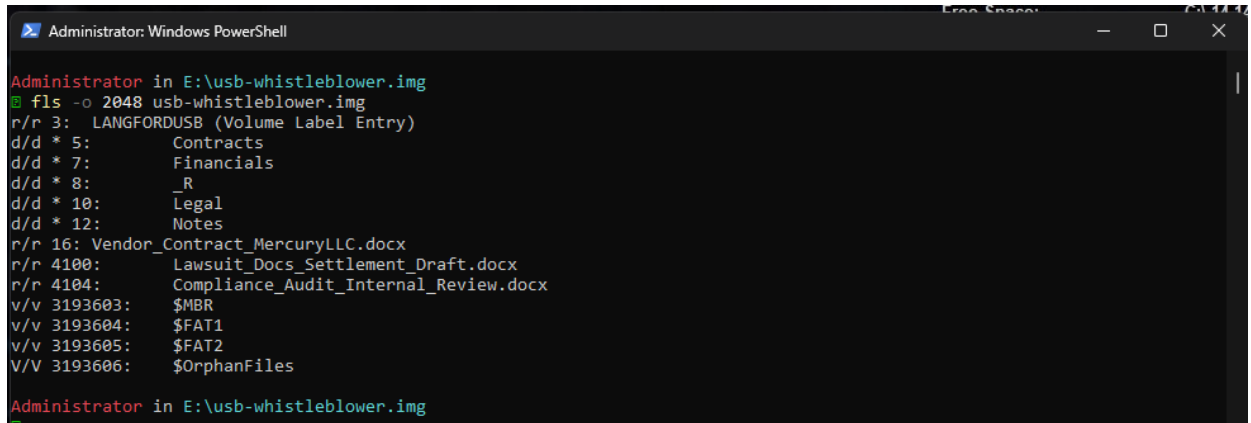
The first 11 bytes of the root directory entry were converted from hex to ASCII in WinHex to spell out 'LANGFORDUSB'.

WinHex - [usb-whistleblower.img]															
File Edit Search Navigation View Tools Specialist Options Window Help															
Case Data															
File Edit															
Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E F
000289FF0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00028A000	4C	41	4E	47	46	4F	52	44	55	53	42	08	00	00	20 6C
00028A010	87	5A	87	5A	00	00	20	6C	87	5A	00	00	00	00	00
00028A020	E5	43	00	6F	00	6E	00	74	00	72	00	0F	00	B4	61 00
00028A030	63	00	74	00	73	00	00	00	FF	FF	00	00	FF	FF	FF FF
00028A040	E5	4F	4E	54	52	41	7E	31	20	20	20	10	00	60	20 6C
00028A050	87	5A	86	5A	00	00	2D	99	84	5A	03	00	00	00	00
00028A060	E5	46	00	69	00	6E	00	61	00	6E	00	0F	00	7C	63 00
00028A070	69	00	61	00	6C	00	73	00	00	00	00	00	FF	FF	FF FF
00028A080	E5	49	4E	41	4E	43	7E	31	20	20	20	10	00	60	20 6C
00028A090	87	5A	86	5A	00	00	35	99	84	5A	4F	00	00	00	00
00028A0A0	E5	52	20	20	20	20	20	20	20	20	20	10	00	61	20 6C
00028A0B0	87	5A	86	5A	00	00	3B	99	84	5A	5D	00	00	00	00
00028A0C0	E5	4C	00	65	00	67	00	61	00	6C	00	0F	00	EB	00 00
00028A0D0	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF FF
00028A0E0	E5	45	47	41	4C	20	20	20	20	20	20	10	00	61	20 6C
00028A0F0	87	5A	86	5A	00	00	43	99	84	5A	6B	00	00	00	00
00028A100	E5	4E	00	6F	00	74	00	65	00	73	00	0F	00	24	00 00
00028A110	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF FF
00028A120	E5	4F	54	45	53	20	20	20	20	20	20	10	00	62	20 6C
00028A130	87	5A	86	5A	00	00	25	99	84	5A	B7	00	00	00	00
00028A140	43	2E	00	64	00	6F	00	63	00	78	00	0F	00	54	00 00
00028A150	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF FF
00028A160	02	63	00	74	00	5F	00	4D	00	65	00	0F	00	54	72 00
00028A170	63	00	75	00	72	00	79	00	4C	00	00	00	4C	00	43 00
00028A180	01	56	00	65	00	6E	00	64	00	6F	00	0F	00	54	72 00
00028A190	5F	00	43	00	6F	00	6E	00	74	00	00	00	72	00	61 00
00028A1A0	56	45	4E	44	4F	52	7E	31	44	4F	43	20	00	64	20 6C
00028A1B0	87	5A	87	5A	00	00	20	6C	87	5A	BA	00	68	8F	00 00
00028A1C0	43	61	00	66	00	74	00	2E	00	64	00	0F	00	41	6F 00
00028A1D0	63	00	78	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF FF
00028A1E0	02	53	00	65	00	74	00	74	00	6C	00	0F	00	41	65 00
00028A1F0	6D	00	65	00	6E	00	74	00	5F	00	00	00	44	00	72 00
00028A200	2E	20	20	20	20	20	20	20	20	20	20	10	00	60	20 6C
00028A210	87	5A	87	5A	00	00	20	6C	87	5A	03	00	00	00	00
00028A220	2E	2E	20	20	20	20	20	20	20	20	20	10	00	60	20 6C
00028A230	87	5A	87	5A	00	00	20	6C	87	5A	00	00	00	00	00
00028A240	E5	41	00	63	00	6D	00	65	00	2E	00	0F	00	A3	70 00
00028A250	64	00	66	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF FF
00028A260	E5	4E	00	44	00	41	00	5F	00	54	00	0F	00	A3	65 00
00028A270	6D	00	70	00	6C	00	61	00	74	00	00	00	65	00	5F 00
00028A280	E5	44	41	5F	54	45	7E	31	50	44	46	20	00	60	20 6C
00028A290	87	5A	86	5A	00	00	2B	53	B7	58	04	00	C4	04	00 00
00028A2A0	E5	2E	00	64	00	6F	00	63	00	78	00	0F	00	54	00 00
00028A2B0	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF FF
00028A2C0	E5	63	00	74	00	5F	00	4D	00	65	00	0F	00	54	72 00
00028A2D0	63	00	75	00	72	00	79	00	4C	00	00	00	4C	00	43 00
00028A2E0	E5	56	00	65	00	6E	00	64	00	6F	00	0F	00	54	72 00
00028A2F0	5F	00	43	00	6F	00	6E	00	74	00	00	00	72	00	61 00
00028A300	E5	45	4E	44	4F	52	7E	31	44	4F	43	20	00	60	20 6C
00028A310	87	5A	87	5A	00	00	A4	A6	C7	58	07	00	68	8F	00 00
00028A320	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00028A330	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Figure 9: WinHex offset at the root directory

7. There are three regular files under the root directory of the file system, what are they?

There are 3 regular files under the root directory: *Vendor_Contract_MercuryLLC.docx*, *Lawsuit_Docs_Settlement_Draft.docx*, and *Compliance_Audit_Internal_Review.docx*. These files were identified by the r/r entries in the 'fls' output.

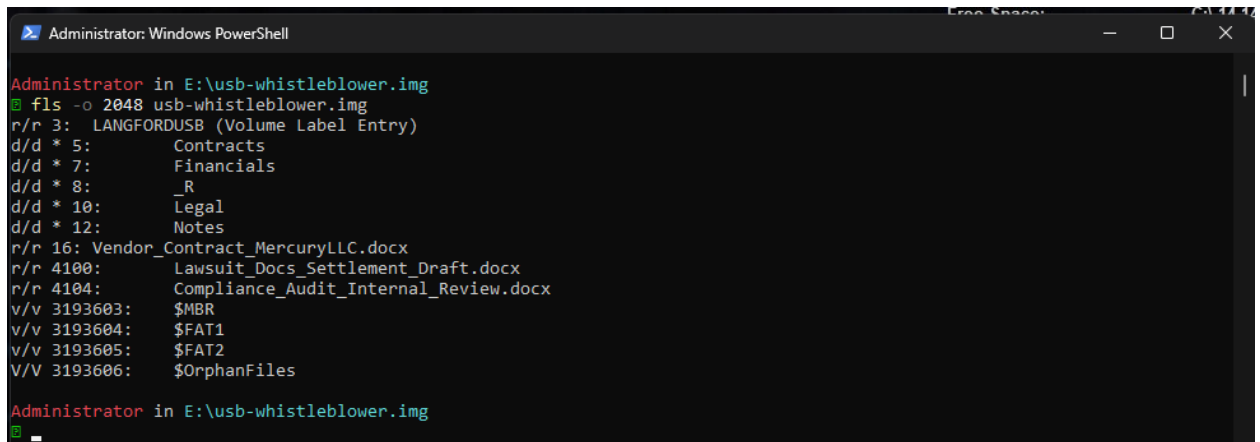


```
Administrator in E:\usb-whistleblower.img
fls -o 2048 usb-whistleblower.img
r/r 3: LANGFORDUSB (Volume Label Entry)
d/d * 5: Contracts
d/d * 7: Financials
d/d * 8: _R
d/d * 10: Legal
d/d * 12: Notes
r/r 16: Vendor_Contract_MercuryLLC.docx
r/r 4100: Lawsuit_Docs_Settlement_Draft.docx
r/r 4104: Compliance_Audit_Internal_Review.docx
v/v 3193603: $MBR
v/v 3193604: $FAT1
v/v 3193605: $FAT2
V/V 3193606: $OrphanFiles
Administrator in E:\usb-whistleblower.img
```

Figure 10: TSK 'fls' command to find files under the root directory

8. Are there any directories under the root directory? Name them.

The directories under the root directory are *Contracts*, *Financials*, *_R*, *Legal*, and *Notes*, as shown by the d/d entries in the 'fls' output.



```
Administrator in E:\usb-whistleblower.img
fls -o 2048 usb-whistleblower.img
r/r 3: LANGFORDUSB (Volume Label Entry)
d/d * 5: Contracts
d/d * 7: Financials
d/d * 8: _R
d/d * 10: Legal
d/d * 12: Notes
r/r 16: Vendor_Contract_MercuryLLC.docx
r/r 4100: Lawsuit_Docs_Settlement_Draft.docx
r/r 4104: Compliance_Audit_Internal_Review.docx
v/v 3193603: $MBR
v/v 3193604: $FAT1
v/v 3193605: $FAT2
V/V 3193606: $OrphanFiles
Administrator in E:\usb-whistleblower.img
```

Figure 11: TSK 'fls' command to find directories under the root directory

9. Are these folders allocated?

These folders are unallocated because the * symbol next to each directory in the 'fls' output indicates that they have been deleted.

```
Administrator: Windows PowerShell
Administrator in E:\usb-whistleblower.img
fls -o 2048 usb-whistleblower.img
r/r 3:  LANGFORDUSB (Volume Label Entry)
d/d * 5:  Contracts
d/d * 7:  Financials
d/d * 8:  _R
d/d * 10: Legal
d/d * 12: Notes
r/r 16: Vendor_Contract_MercuryLLC.docx
r/r 4100: Lawsuit_Docs_Settlement_Draft.docx
r/r 4104: Compliance_Audit_Internal_Review.docx
v/v 3193603: $MBR
v/v 3193604: $FAT1
v/v 3193605: $FAT2
V/V 3193606: $OrphanFiles

Administrator in E:\usb-whistleblower.img
_
```

Figure 12: TSK 'fls' command to find directories under the root directory

10. List all unallocated directories?

The unallocated directories are *Contracts*, *Financials*, *_R*, *Legal*, and *Notes*, as identified by the 'fls -d' output which lists deleted directory entries.

```
Administrator: Windows PowerShell
V/V 3193606: $OrphanFiles

Administrator in E:\usb-whistleblower.img
fls -d -o 2048 usb-whistleblower.img
d/d * 5:  Contracts
d/d * 7:  Financials
d/d * 8:  _R
d/d * 10: Legal
d/d * 12: Notes

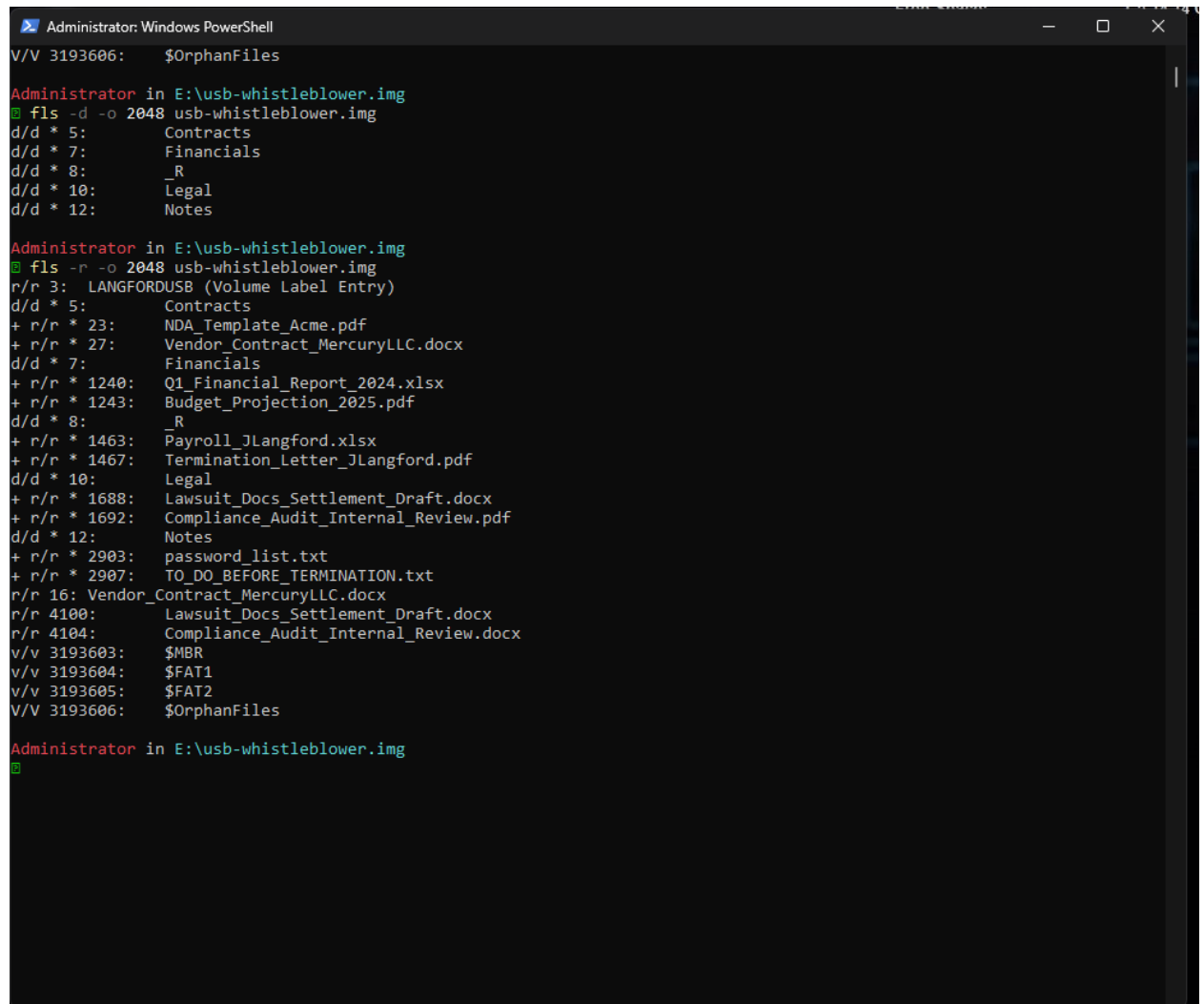
Administrator in E:\usb-whistleblower.img
_
```

Figure 13: TSK 'fls -d' command to show deleted directory entries

11. List all unallocated files?

The unallocated files are *NDA_Template_Acme.pdf*, *Vendor_Contract_MercuryLLC.docx*, *Q1_Financial_Report_2024.xlsx*, *Budet_Projection_2025.pdf*, *Payroll_JLangford.xlsx*, *Termination_Letter_JLangford.pdf*, *Lawsuit_Docs_Settlement_Draft.docx*, *Compliance_Audit_Internal_Review.pdf*, *password_list.txt*, and

TO_DO_BEFORE_TERMINATION.txt. We can see this because the * symbol in the recursive 'fls' output marks them as deleted/unallocated.



```
Administrator: Windows PowerShell
V/V 3193606: $OrphanFiles

Administrator in E:\usb-whistleblower.img
fls -d -o 2048 usb-whistleblower.img
d/d * 5: Contracts
d/d * 7: Financials
d/d * 8: _R
d/d * 10: Legal
d/d * 12: Notes

Administrator in E:\usb-whistleblower.img
fls -r -o 2048 usb-whistleblower.img
r/r 3: LANGFORDUSB (Volume Label Entry)
d/d * 5: Contracts
+ r/r * 23: NDA_Template_Acme.pdf
+ r/r * 27: Vendor_Contract_MercuryLLC.docx
d/d * 7: Financials
+ r/r * 1240: Q1_Financial_Report_2024.xlsx
+ r/r * 1243: Budget_Projection_2025.pdf
d/d * 8: _R
+ r/r * 1463: Payroll_JLangford.xlsx
+ r/r * 1467: Termination_Letter_JLangford.pdf
d/d * 10: Legal
+ r/r * 1688: Lawsuit_Docs_Settlement_Draft.docx
+ r/r * 1692: Compliance_Audit_Internal_Review.pdf
d/d * 12: Notes
+ r/r * 2903: password_list.txt
+ r/r * 2907: TO_DO_BEFORE_TERMINATION.txt
r/r 16: Vendor_Contract_MercuryLLC.docx
r/r 4100: Lawsuit_Docs_Settlement_Draft.docx
r/r 4104: Compliance_Audit_Internal_Review.docx
v/v 3193603: $MBR
v/v 3193604: $FAT1
v/v 3193605: $FAT2
V/V 3193606: $OrphanFiles

Administrator in E:\usb-whistleblower.img
```

Figure 14: TSK 'fls -r' command to show unallocated files

12. Are any unallocated files you just mentions located elsewhere on this disk? List them.

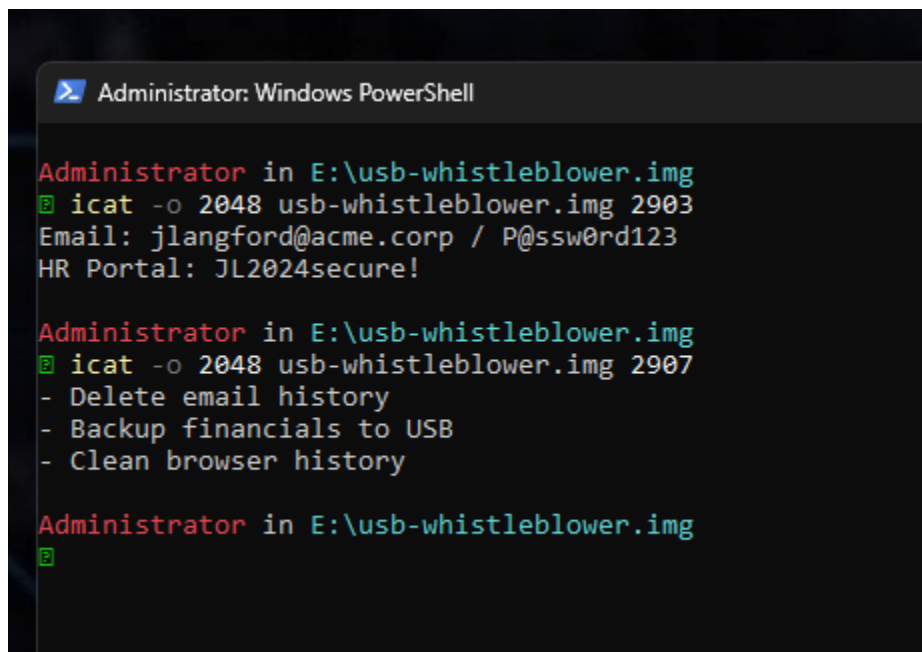
Yes, unallocated files are located elsewhere on the disk because the recursive 'fls -r' output shows multiple deleted files (marked with *) within deleted directories, indicating that unallocated file data still exists on the disk. The unallocated files are *NDA_Template_Acme.pdf*, *Vendor_Contract_MercuryLLC.docx*, *Q1_Financial_Report_2024.xlsx*, *Budet_Projection_2025.pdf*, *Payroll_Jlangford.xlsx*, *Termination_Letter_JLangford.pdf*, *Lawsuit_Docs_Settlement_Draft.docx*, *Compliance_Audit_Internal_Review.pdf*, *password_list.txt*, and *TO_DO_BEFORE_TERMINATION.txt*.

13. Are there any files that may give us insight into the motives or methods of Ms. Langford? List the file names?

Files that may provide insight into Ms. Langford's motives or methods include *password_list.txt* and *TO_DO_BEFORE_TERMINATION.txt*, as well as documents such as *Termination_Letter_JLangford.pdf*, *Payroll_JLangford.xlsx*, and *Lawsuit_Docs_Settlement_Draft.docx*, because these files suggest preparation prior to termination, access to sensitive company data, and potential intent to retain or misuse that information.

14. What is the content of this file?

The file that may contain the most important information is the *password_list.txt* and *TO_DO_BEFORE_TERMINATION.txt*. These files suggest that Ms. Langford maintained access credentials that could have been used to access or transfer company data after her termination. The contents of the *password_lists.txt* was Ms. Langford's email address and password to her work account. The content of the *TO_DO_BEFORE_TERMINATION.txt* was a list of offboarding instructions.



```
Administrator: Windows PowerShell

Administrator in E:\usb-whistleblower.img
E icat -o 2048 usb-whistleblower.img 2903
Email: jlangford@acme.corp / P@ssw0rd123
HR Portal: JL2024secure!

Administrator in E:\usb-whistleblower.img
E icat -o 2048 usb-whistleblower.img 2907
- Delete email history
- Backup financials to USB
- Clean browser history

Administrator in E:\usb-whistleblower.img
E
```

15. Another team is reporting that Ms. Lanford encrypted her laptop with a password. Can you give them any information that would help their investigation?

The file *password_list.txt* may assist in accessing the encrypted laptop because it contains a stored password and credential information, from Ms. Langford, in plaintext. This password could be used to unlock the device or any of her accounts. The presence of credential files and pre-termination planning documents suggests that Ms. Langford may have intentionally prepared to retain access to company data and remove sensitive information prior to her termination.

Conclusion

The forensic examination of the USB thumb drive revealed critical evidence regarding the structure, contents, and potential misuse of company data by Ms. Jessica Langford. The analysis confirmed that the device utilized a GPT partitioning scheme with a FAT32 file system and contained both allocated and unallocated data. Through the use of The Sleuth Kit and WinHex, multiple files and directories were identified, including several deleted directories and recoverable files containing sensitive corporate information.

Notably, the presence of deleted directories alongside recoverable files suggests an attempt to conceal data rather than securely delete it. Additionally, the discovery of files such as *password_list.txt* and *TO_DO_BEFORE_TERMINATION.txt* indicates possible premeditation and provides insight into Ms. Langford's methods and intent. These findings suggest that Ms. Langford may have retained access credentials and organized company data prior to her termination.

Overall, this investigation demonstrates a thorough forensic process, including disk analysis, file system examination, and recovery of deleted artifacts. The evidence identified on the USB device supports the possibility of data exfiltration and intentional actions taken before termination, providing valuable information for further legal and investigative efforts.

References

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“Help Documents.” *The Sleuth Kit Wiki*, 2026, wiki.sleuthkit.org/Help-Documents/. Accessed 23 Apr. 2026.